Decoupling What from Where: A Mechanism Study of Action-Type Supervision for GUI Grounding in a Small Vision-Language Model

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Abstract

A GUI agent decides which action to take and where on the screen to take it; native agent models emit both in one token stream. We ask what supervising the action type does for the grounding half of that decision, using Qwen2-VL-2B with LoRA on Android in the Wild. A flat baseline is compared with five ways of supplying the type under matched data, compute, adapters, and decoding: an auxiliary loss, a hard-routed action word, an additive learned embedding, a prepended learned token, and the type written into the prompt. With five seeds, a cluster bootstrap over validation examples, and seed-level paired tests, two results hold. On a training stream that mixes clicks and scrolls with type events whose recorded target is a fixed off-screen point, the auxiliary loss, the additive embedding, and the type written into the prompt each raise click grounding by 6 to 10 points over the baseline; the prepended learned token, the mechanism we originally hypothesized, is no better than the baseline. The benefit is largely protection from the degenerate class: removing those events from training raises the baseline itself by four points, after which no mechanism beats it on the same validation examples, and on a separate stream of taps and swipes no mechanism helps either. Interventions on the trained models show that both learned embeddings are consulted at inference, with a difference that matters for deployment: with its embedding zeroed, the additive model returns to the baseline's accuracy, whereas the prepended-token model falls below it. We also correct our own description of a silent failure: injecting the conditioning through `inputs_embeds` makes Qwen2-VL fall back to one-dimensional positions for image tokens, which produced a nine-point deficit that vanished once `input_ids` were preserved.

1 Introduction

A GUI agent maps a screenshot and an instruction to a low-level action: an action type such as *click*, *scroll*, or *type*, and, for spatial actions, a target location. Current native agent models (UI-TARS [Qin et al., 2025], OS-Atlas [Wu et al., 2025c], SeeClick [Cheng et al., 2024], ShowUI [Lin et al., 2025]) produce both in a single autoregressive stream. The two decisions are different kinds of prediction: the action type is a small classification problem that depends on the goal and the screen state, and the location is a localization problem in which the model has to find the evidence on the screen that supports the action and commit to a point. Several recent agents separate the two, predicting the type first and the target conditioned on it [Ma et al., 2024, Christianos et al., 2025]. What has not been measured is what the type supervision itself does to the grounding model, and which of the many ways of delivering it matters.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

This paper is a matched-compute study of that question in a small model. We fine-tune Qwen2-VL-2B [Wang et al., 2024] with LoRA [Hu et al., 2022] to emit a coordinate string, and compare a flat baseline with five conditioning mechanisms that receive the action type in different forms: as an auxiliary classification loss, as a hard-routed action word in the output, as an additive learned embedding, as a prepended learned token, and as a word in the prompt. Each variant sees the same data in the same order and uses the same optimizer, adapters, and decoding. We started from the hypothesis that the prepended token would be the right mechanism, because it gives the model a dedicated, attendable position for the action type.

The results support type supervision and refute our hypothesis about the mechanism, and they depend on a property of the data that we did not appreciate at first. Android in the Wild records type events with a touch point off the screen, which our coordinate serializer clamps to the origin, so a training stream that mixes clicks, scrolls, and type events contains a class whose target is a fixed point. On that stream, the auxiliary loss and the additive embedding raise hit@0.10 by $+0.064$ and $+0.054$ over the flat baseline (five seeds, cluster bootstrap over examples, seed-level $p = 0.004$ and 0.028), almost entirely on clicks; the type written into the prompt as a word, with no new parameters, does as well, and the prepended token is within noise of the baseline. The obvious explanation, that the baseline learns to emit the off-screen point for clicks, accounts for little of the gap (it does so on 4% of clicks), but the class does hurt the baseline more broadly: removing type events from training while scoring the identical validation examples raises the baseline by four points, after which no mechanism beats it, and on a separate stream of taps and swipes none helps either.

Interventions on the trained models explain the ranking. A wrong type collapses either embedding model and a zeroed or class-averaged embedding costs accuracy in both, so both learned embeddings are consulted at inference; the difference is what is left when the signal is removed. The zeroed additive model sits at the flat baseline's level, the zeroed prepended-token model seven points below it. An earlier implementation that injected the conditioning through `inputs_embeds`, which makes Qwen2-VL fall back to one-dimensional positions for its image tokens, scored nine points below the baseline, a result we initially read as evidence against conditioning.

Our contributions are (1) a matched-compute comparison of five action-type conditioning mechanisms against a flat baseline, with five seeds, a cluster bootstrap over examples, seed-level tests, and a control stream where conditioning should not help; (2) a diagnosis of where the gain comes from, including retraining with the degenerate class removed; (3) interventions that separate what each mechanism does at inference from what it does in training; and (4) a corrected account of a positional-encoding failure that inverted a conclusion.

2 Related Work

GUI grounding and agent models. Mind2Web [Deng et al., 2023] and Android in the Wild [Rawles et al., 2023] supply the web and mobile episodes we train on; AndroidControl [Li et al., 2024] and AndroidWorld [Rawles et al., 2025] extend mobile evaluation. SeeClick [Cheng et al., 2024] argued that grounding is the bottleneck for visual GUI agents and introduced ScreenSpot, which ScreenSpot-Pro [Li et al., 2025] extends. UGround [Gou et al., 2025] and OS-Atlas [Wu et al., 2025c] scale grounding pretraining; Aguvis [Xu et al., 2025], UI-TARS [Qin et al., 2025], CogAgent [Hong et al., 2024], ScreenAI [Baechler et al., 2024], and Ferret-UI [You et al., 2024b] emit action and location together, the design our flat baseline follows. GUI-Actor [Wu et al., 2025a] replaces coordinate text with an attention head, UI-R1 [Lu et al., 2026] rewards type and argument separately, RegionFocus [Luo et al., 2025] zooms at test time, and OmniParser [Lu et al., 2024] and Set-of-Mark [Yang et al., 2023] externalize grounding into a parser.

Type-first agents. CoCo-Agent [Ma et al., 2024] decomposes AITW action prediction into the type followed by the target conditioned on it, in one stream; that is our hard-routing variant. Li-MAC [Christianos et al., 2025] places a small action-type transformer ahead of a LoRA-tuned Qwen2-VL-2B on AITW and AndroidControl and selects click targets conditioned on the predicted type, using screen elements rather than pixels. SeeAct [Zheng et al., 2024], Ponder & Press [Wang et al., 2025], and Aria-UI [Yang et al., 2025] separate text planning from pixel grounding; GUI-Libra [Yang et al., 2026] reweights reasoning against action tokens and GUI-AIMA [Zhou et al., 2025] adds an anchor token for grounding. These systems establish that type-first pipelines work;

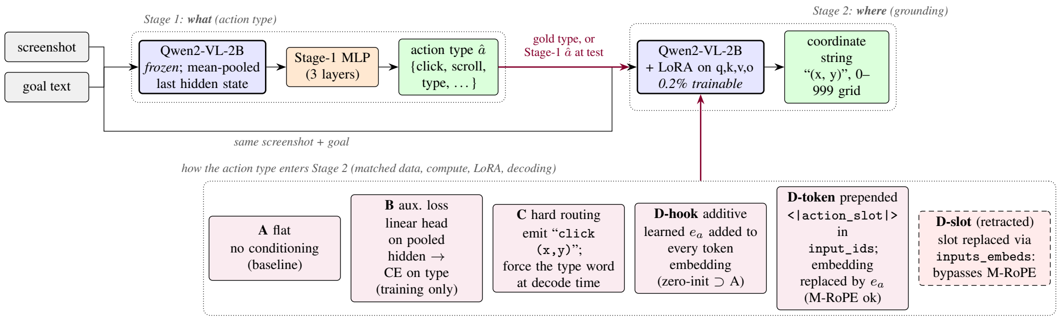

Figure 1: Two-stage pipeline. Stage 1 classifies the action type from mean-pooled frozen features. Stage 2 grounds the action to a coordinate string, and the action type enters it through one of the mechanisms in the bottom row (D-text, the type as a word in the prompt, is not drawn). All mechanisms share data, compute, adapters, and decoding. D-slot is the retracted variant whose `inputs_embeds` injection bypassed the position computation.

none holds data, compute, and decoding fixed across conditioning mechanisms or intervenes on the trained model to ask whether the type signal is used, which is what this paper does.

Conditioning, faithfulness, and the backbone. SayCan [Ahn et al., 2022], ReAct [Yao et al., 2023], and RT-2 [Brohan et al., 2023] are the reference points for factored, exposed, and token-level action decisions; AutoUI [Zhang and Zhang, 2024], AppAgent [Zhang et al., 2025], DigiRL [Bai et al., 2024], WebRL [Qi et al., 2025], and VSC-RL [Wu et al., 2025b] improve mobile agents by imitation or reinforcement learning without changing how the type enters the grounding computation. Our auxiliary-loss variant is a multi-task objective in the sense of Caruana [1997]. Kosmos-2 [Peng et al., 2024], Shikra [Chen et al., 2023], and Ferret [You et al., 2024a] train VLMs to refer to image regions, and POPE [Li et al., 2023] and HallusionBench [Guan et al., 2024] ask whether an output is supported by the image; our attention measurements ask that of a coordinate. Qwen2-VL [Wang et al., 2024] assigns three-dimensional rotary positions (extending RoPE [Su et al., 2024]) to visual tokens from `input_ids`; Section 6.2 describes what happens when conditioning bypasses that path.

3 Method

3.1 Action taxonomy and grounding target

Both datasets are mapped onto a canonical eight-class action space (*click*, *double-click*, *type*, *scroll*, *drag*, *hotkey*, *wait*, *finished*). Mind2Web's CLICK, SELECT, and HOVER map to *click* and TYPE to *type*. AITW stores gestures as touch and lift points; we split them into taps (*click*) and swipes (*scroll*) by the normalized distance between the two, with threshold 0.04. The grounding target is the touch point normalized to $[0, 1]^2$, serialized as `(x, y)` with integers on a 0–999 grid so every target has the same token length. AITW records non-touch actions such as typing with the touch point $(-1, -1)$; the serializer clamps this to `(0, 0)`, so every *type* event in the training stream has the same target at the screen origin, and its distance to the raw target is exactly $\sqrt{2}$ whatever the model predicts. This detail matters for the analysis in Section 5.1.

3.2 Stage 1: which action

Stage 1 runs the frozen Qwen2-VL-2B-Instruct backbone over the screenshot and instruction, mean-pools the final hidden states over unmasked tokens into a 1536-dimensional vector, and trains a three-layer MLP ($1536 \to 1024 \to 256 \to 8$, dropout 0.1) with class-weighted cross-entropy, AdamW (learning rate 10^{-4}, weight decay 0.01), batch size 128, for eight epochs, keeping the checkpoint with the best validation macro-F1. We compare features from text only, text plus screenshot, and a sanity variant with zeroed features.

3.3 Stage 2: where

Stage 2 fine-tunes the same backbone with LoRA adapters (rank 16, $\alpha = 32$, dropout 0.05) on the query, key, value, and output projections, 4.4M trainable parameters out of 2.2B. The prompt is the screenshot, `Goal: {goal}`, and a fixed instruction to predict the action coordinate; the answer is the coordinate string. With s the screenshot, g the goal, a the action type, and $y_{1:T}$ the coordinate tokens, every variant minimizes

$$\mathcal{L}_{\text{coord}} = -\sum_{t=1}^{T} \log p_\theta\big(y_t \mid y_{<t},\, s,\, g,\, c\big), \tag{1}$$

masked to the answer tokens, with c the variant-specific conditioning.

A, flat. $c = \varnothing$: the joint-decoding design of native agent models, reduced to the grounding sub-problem.

B, auxiliary loss. A linear head h_ϕ on the mean-pooled last hidden state $\bar{z}$ predicts the action type during training, $\mathcal{L}_{\text{B}} = \mathcal{L}_{\text{coord}} + \lambda\,\text{CE}(h_\phi(\bar{z}), a)$ with $\lambda = 1$. Nothing about the type is provided at inference, so any gain is a training effect.

C, hard routing. The model is trained to emit the action word before the coordinate, `click (x, y)`, as in CoCo-Agent's conditional action prediction. At test time the word for the gold (or Stage-1) type is forced as a decoding prefix.

D-hook, additive embedding. A learned table $E \in \mathbb{R}^{8\times d}$ holds one vector per class; a forward hook on the embedding layer adds $E[a]$ to every text and image token embedding. E is zero-initialized, so D-hook starts as an exact copy of A.

D-token, prepended action token. A new token `<|action_slot|>` is added to the vocabulary and placed at the start of the text prompt, so it occupies a real position in `input_ids`; a forward hook replaces its embedding with $E[a]$, initialized from $\mathcal{N}(0, 0.02^2)$. This is the architecture we originally hypothesized: a full-norm, attendable representation of the type.

D-text, type as a word. The prompt begins with `Action: click.` (or the gold type's word), using the pretrained word embeddings and no new parameters. This is the obvious baseline for the two learned-embedding variants.

For C, D-hook, D-token, and D-text the gold type is used in training and at evaluation unless stated otherwise; Section 5.3 closes the loop with Stage-1 predictions. All variants share the data order, the optimizer (AdamW, learning rate 2×10^{-5}, weight decay 0.01, batch size 1, gradient clipping at 1.0, two epochs), and greedy decoding with 16 new tokens. No hyperparameter was tuned per variant; Section 7 returns to this.

4 Experimental setup

Data. Stage 1 uses 7,775 Multimodal-Mind2Web training steps evaluated on the 1,338-step `test_task` split, and 5,000 AITW steps from the General subset evaluated on 1,000 held-out steps. Stage 2 uses two AITW streams built by filtering the training split in its stored order. `all_with_coords` keeps taps, swipes, and type events; at 1,200 training examples its validation slice of 250 steps holds 169 clicks, 46 scrolls, and 35 type events. `taps_and_swipes` keeps only taps and swipes (1,000 training and 200 validation steps); both land anywhere on the screen. The AITW stream is deterministic, so for a given mix and training size every seed and every variant is evaluated on the same examples in the same order, which is what makes paired tests possible. The validation slice begins where the training slice ends; consecutive steps come from the same episodes, and absolute numbers are not comparable across training sizes. A second control uses Multimodal-Mind2Web (1,000 training and 250 evaluation steps), where the target is the center of the gold element. The Stage-2 study is run on AITW because that is where the action type is a vision problem: Stage 1 (Table 5 in the appendix) gains $+0.414$ macro-F1 from the screenshot on AITW, whose instructions are goals, and $+0.009$ on Mind2Web, whose instructions name the action.

Metrics. hit@r is the fraction of predictions within normalized Euclidean distance r of the target, for $r \in \{0.05, 0.10, 0.25\}$; hit@0.10 is the headline. We also report the mean normalized distance.

Table 1: Stage 2 grounding on AITW `all_with_coords` (n_{train}=1200, n_{val}=250, 5 seeds). Mean ± std over seeds; Δ vs. A with a 95% cluster-bootstrap CI over validation examples (all seeds of an example kept together; stars from the same bootstrap, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$) and the seed-level paired t-test p.

Variant	hit@0.05	hit@0.10	hit@0.25	mean L2 ↓	Δhit@0.10 vs. A [cluster CI]	seed-level p
A: flat	0.130 ± 0.039	0.229 ± 0.043	0.499 ± 0.029	0.406 ± 0.028		
B: aux. loss	0.178 ± 0.025	0.293 ± 0.021	0.582 ± 0.026	0.365 ± 0.004	+0.064 [+0.036, +0.093]***	0.004
C: hard routing	0.170 ± 0.022	0.264 ± 0.045	0.538 ± 0.059	0.421 ± 0.016	+0.035 [+0.001, +0.069]*	0.189
D-hook: additive	0.170 ± 0.042	0.282 ± 0.041	0.563 ± 0.016	0.370 ± 0.010	+0.054 [+0.022, +0.086]***	0.028
D-token: prepended	0.144 ± 0.056	0.238 ± 0.049	0.504 ± 0.055	0.386 ± 0.021	+0.009 [-0.015, +0.034]	0.465
D-text: word in prompt	0.186 ± 0.022	0.302 ± 0.028	0.568 ± 0.033	0.364 ± 0.008	+0.073 [+0.033, +0.113]***	0.018

Unparseable outputs count as misses at distance $\sqrt{2}$; only hard routing produces them (2 to 8% of outputs), and its numbers are reported both ways. Per-class numbers are computed on the same basis as the overall ones.

Statistics. The headline setting uses five seeds (42 to 46); every other cell uses three (42 to 44). We report the mean and sample standard deviation over seeds. Every run logs a per-example distance on a shared evaluation set, and every seed of every variant shares the per-seed data order, so comparisons are paired at two levels. Our primary interval is a cluster bootstrap that resamples validation examples with replacement and keeps all seeds of a sampled example together (10,000 resamples, 95% percentile interval, two-sided bootstrap p [Efron and Tibshirani, 1993]); resampling (seed, example) units separately would treat the seeds of one example as independent and gives intervals about a third narrower. We also report a paired t-test on the five (or three) per-seed means, which has little power but cannot be fooled by within-seed correlation, and treat a difference as established when both agree; stars come from the cluster bootstrap with the seed-level p beside them. No correction is made for the number of contrasts; the scaling study in Section 5.4 is presented as descriptive for that reason.

5 Results

5.1 Type supervision improves click grounding on the mixed stream

Table 1 is the main result: six variants on `all_with_coords` at 1,200 training examples, five seeds each. The auxiliary loss (B) and the additive embedding (D-hook) beat the flat baseline on every metric, by $+0.064$ and $+0.054$ hit@0.10 with cluster intervals that exclude zero and seed-level p of 0.004 and 0.028, and they are indistinguishable from each other (D-hook − B: -0.010, cluster CI $[-0.043, +0.024]$, seed $p = 0.56$). Hard routing (C) gains $+0.035$ with a cluster interval that barely excludes zero and a seed-level p of 0.19, and loses on mean distance, partly because forcing the action word makes 2 to 8% of its outputs unparseable; scored on parsed outputs only, its click rate is unchanged (0.356 vs. 0.350). The prepended token (D-token), our hypothesized mechanism, is within noise of the baseline on every hit rate ($+0.009$, cluster CI $[-0.015, +0.034]$) and below B by -0.055 (cluster CI $[-0.088, -0.024]$, seed $p = 0.016$). The type written into the prompt as a word (D-text), which adds no parameters, gains $+0.073$ over the baseline (cluster CI $[+0.033, +0.113]$, seed $p = 0.018$) and is within noise of B ($+0.009$, cluster CI $[-0.036, +0.053]$), so the failure is specific to the learned prepended token, not to conditioning at the input. Seed variance is large: the baseline ranges from 0.169 to 0.280 across seeds, which is why we report five seeds and paired statistics rather than a best run; the ordering with the first three seeds was the same.

Table 2 shows where the gain lives and tests the simplest explanation for it. The improvement is a click improvement: B and C gain ten points on clicks, D-hook six, and C pays for its click gain with a sixteen-point loss on scrolls because forcing the word *scroll* into the output derails the gesture; D-hook is the only variant that improves both classes. Since the only difference between this stream and the control is the type class with its fixed target, one might expect the flat baseline's click errors to be emissions of that target that the conditioned models avoid. They mostly are not. The baseline emits `(0, 0)` for 4.3% of clicks (13% on its worst seed), the auxiliary-loss model for 2.4%, and the three conditioned-at-inference models never; excluding those predictions moves the baseline's click rate from 0.256 to 0.266 while B stays at 0.366 and D-hook at 0.321. The class degrades the baseline's click grounding more broadly than by exact emissions, and Table 7 in the

Table 2: Where the gain lives. Per-class hit@0.10 on the headline mix (five seeds, same basis as Table 1), the fraction of click predictions that are the clamped type target `(0, 0)`, and click hit@0.10 after excluding those predictions. Every model scores zero on *type* because the raw target is off-screen.

Variant	click	scroll	type	clicks sent to `(0, 0)`	click, excluding those
A: flat	0.256 ± 0.062	0.304 ± 0.034	0.000	4.3% (13.0% on the worst seed)	0.266
B: aux. loss	0.357 ± 0.026	0.278 ± 0.018	0.000	2.4%	0.366
C: hard routing	0.350 ± 0.061	0.148 ± 0.073	0.000	0.0%	0.350
D-hook: additive	0.321 ± 0.064	0.357 ± 0.039	0.000	0.0%	0.321
D-token: prepended	0.269 ± 0.069	0.304 ± 0.090	0.000	0.0%	0.269

Table 3: Control: AITW `taps_and_swipes` (n_{train}=1000, n_{val}=200, 3 seeds), a training stream without the degenerate class. Same statistics as Table 1.

Variant	hit@0.05	hit@0.10	hit@0.25	mean L2 ↓	Δhit@0.10 vs. A [cluster CI]	seed-level p
A: flat	0.243 ± 0.020	0.382 ± 0.015	0.720 ± 0.017	0.185 ± 0.003		
B: aux. loss	0.225 ± 0.015	0.368 ± 0.012	0.730 ± 0.039	0.173 ± 0.011	-0.013 [-0.038, +0.012]	0.456
C: hard routing	0.217 ± 0.016	0.325 ± 0.040	0.635 ± 0.064	0.271 ± 0.068	-0.057 [-0.108, -0.007]*	0.215
D-hook: additive	0.235 ± 0.022	0.363 ± 0.021	0.725 ± 0.043	0.182 ± 0.009	-0.018 [-0.052, +0.015]	0.334

appendix measures how much: we retrain A, B, and D-hook on the same training slice with the type events removed and score the identical 250 validation examples (three seeds). Removing the class raises the flat baseline from 0.255 to 0.297 hit@0.10 (cluster CI on the change $[+0.016, +0.069]$; clicks from 0.294 to 0.365) and leaves the conditioned models where they were. On that type-free stream the additive embedding's advantage over the stronger baseline vanishes ($+0.008$, cluster CI $[-0.020, +0.035]$; on clicks -0.008), and the auxiliary loss keeps only a small click advantage ($+0.039$, cluster CI $[+0.004, +0.075]$) and a hit@0.25 advantage ($+0.053$, CI $[+0.024, +0.083]$) with no overall hit@0.10 gain ($+0.021$, CI $[-0.004, +0.045]$). With the control in the next section, we read this as follows: what type supervision buys on the mixed stream is protection of click grounding from a class whose target is degenerate, and we do not claim a general spatial prior.

5.2 Control: no gain on taps and swipes

On `taps_and_swipes` (Table 3), whose training stream has no degenerate class, B and D-hook are within noise of the baseline (-0.013 and -0.018 hit@0.10, cluster intervals $[-0.038, +0.012]$ and $[-0.052, +0.015]$), so the data admit harms of up to four points as well as no effect, and C is worse by -0.057 (cluster CI $[-0.108, -0.007]$) with a large increase in mean distance. The Multimodal-Mind2Web control sits at floor (a tenth of predicted points fall inside the gold element for both A and D-hook) and is uninformative; it is reported in the appendix and not leaned on.

5.3 End to end: predicted types instead of gold

The tables above condition on the gold action type. To close the loop we train D-hook, fit the Stage-1 MLP on frozen features of its training examples for a fixed 200 full-batch epochs with no checkpoint selection, and evaluate D-hook on the same validation examples with gold types and with Stage-1 predictions (three seeds; Stage 1 reaches 0.843 accuracy). The oracle pipeline scores 0.292 hit@0.10, the predicted-type pipeline 0.271, and the flat baseline on the same seeds 0.255. Only the oracle-to-predicted gap is established ($+0.021$, cluster CI $[+0.003, +0.041]$); the predicted pipeline's margin over the baseline ($+0.016$, cluster CI $[-0.013, +0.047]$, seed $p = 0.44$) is not, and we report it as descriptive. Classifier error removes more than half of the conditioning gain in this setting.

5.4 Scaling: a descriptive study

Figure 2 extends the comparison to $n \in \{300, 500, 800, 1200, 2500, 5000\}$ with three seeds at every size. The cluster intervals for D-hook $-$ A exclude zero at five of the six sizes on hit@0.10 (all six on hit@0.25), and its estimate at 5,000 examples ($+0.043$) is the same size as at 1,200; B $-$ A excludes zero at 300 and 1,200, is negative at 800, and is within noise elsewhere. With three seeds the seed-level tests reach $p < 0.05$ only at 300 and 5,000 for D-hook and at 300 and 1,200 for B, and a direct D-hook $-$ B contrast is within noise at four of the six sizes. We read the study as consistent

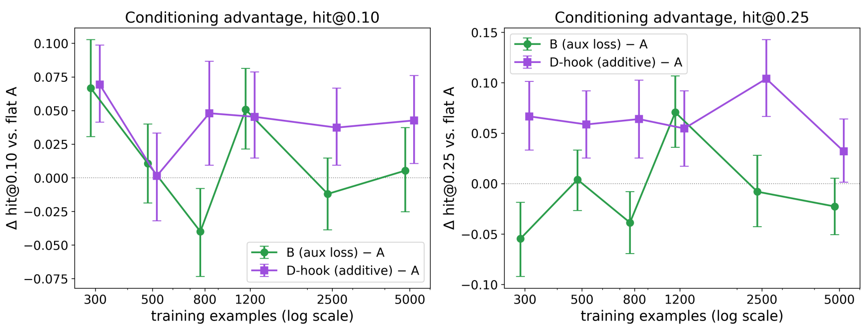

Figure 2: Conditioning advantage over the flat baseline as a function of training-set size on `all_with_coords`, three seeds per cell, 95% cluster-bootstrap intervals over examples. Each size is scored on a different validation slice with a different class mix (Table 8), so only within-size differences are meaningful, and 24 contrasts are shown without correction.

Table 4: Interventions on the trained conditioned models: the same model is decoded with the gold action embedding, a wrong one (cyclic permutation; click→type), a wrong one that swaps only clicks and scrolls, the table zeroed, and every row replaced by the class mean (uninformative, in-distribution norm). Five seeds $\times$ 250 examples; Δ = gold $-$ condition with 95% cluster-bootstrap CI over examples and the seed-level paired p. Flat A on the same examples is the reference row.

Model	Conditioning	hit@0.10	click hit@0.10	scroll hit@0.10	Δ gold $-$ cond. (hit@0.10)	seed p
D-hook: additive	gold type	0.275 ± 0.040	0.310 ± 0.060	0.357 ± 0.052		
	wrong (cyclic)	0.054 ± 0.009	0.006 ± 0.000	0.270 ± 0.050	+0.222 [+0.174, +0.270]***	0.000
	wrong (click↔scroll)	0.099 ± 0.056	0.073 ± 0.087	0.270 ± 0.050	+0.176 [+0.133, +0.221]***	0.003
	zeroed	0.248 ± 0.032	0.288 ± 0.053	0.291 ± 0.050	+0.027 [-0.006, +0.060]	0.224
	class mean	0.206 ± 0.058	0.217 ± 0.099	0.322 ± 0.064	+0.070 [+0.034, +0.106]***	0.133
	flat A, same examples	0.229 ± 0.043	0.256	0.304	+0.046 [+0.017, +0.077]**	0.032
D-token: prepended	gold type	0.242 ± 0.054	0.272 ± 0.079	0.313 ± 0.099		
	wrong (cyclic)	0.079 ± 0.012	0.036 ± 0.019	0.300 ± 0.054	+0.162 [+0.120, +0.205]***	0.001
	wrong (click↔scroll)	0.155 ± 0.043	0.148 ± 0.063	0.300 ± 0.054	+0.086 [+0.043, +0.130]***	0.073
	zeroed	0.162 ± 0.072	0.160 ± 0.112	0.291 ± 0.033	+0.080 [+0.050, +0.112]***	0.120
	class mean	0.169 ± 0.043	0.169 ± 0.067	0.296 ± 0.025	+0.073 [+0.040, +0.107]***	0.091
	flat A, same examples	0.229 ± 0.043	0.256	0.304	+0.013 [-0.010, +0.037]	0.374

with the additive embedding's advantage persisting with data and the auxiliary loss's not, and no stronger than that; an earlier single-seed curve had suggested that all conditioning gains shrink with data, which three seeds do not support. Figure 4 in the appendix shows individual screens.

6 Mechanism

6.1 Interventions on both embedding mechanisms

D-token's failure to help could mean that the model ignores the slot, and D-hook's success could be a training effect that leaves a vector the model never reads. To test both on equal terms we retrain each at the headline setting (five seeds) and decode every trained model under five conditionings on the full validation slice: the gold type; a wrong type from a cyclic permutation of the classes present (click to type, type to scroll, scroll to click); a second wrong map that swaps only clicks and scrolls, so no groundable example is routed to the degenerate class; the table zeroed; and every row replaced by the mean of the rows of the classes present, which is uninformative but in-distribution. Table 4 reports the results with the flat baseline on the same examples; an earlier three-condition test on separately trained D-token models (Table 9, appendix) agrees. The learned rows of D-hook are small: each has norm 0.04 against a mean token-embedding norm of 0.58, whereas D-token's rows have norm 0.78. Small as they are, the model reads them. A wrong type collapses D-hook from 0.275 to 0.054 under the cyclic map and to 0.099 under the click-scroll map (cluster intervals $[+0.174, +0.270]$ and $[+0.133, +0.221]$, seed $p < 0.001$ and 0.003), so the collapse is not an artifact of routing clicks to the degenerate class. The class-mean vector costs seven points ($+0.070$, cluster CI $[+0.034, +0.106]$, seed $p = 0.13$). Zeroing costs about three, and that difference is not established ($+0.027$, cluster CI $[-0.006, +0.060]$, seed $p = 0.22$): the zeroed model sits at the level of the flat baseline ($+0.019$ over A, not established). The single-seed attention run in Appendix G, on which an earlier version of this paper based the claim that the embedding could be removed without loss, is consistent with these intervals and was underpowered. The additive embedding therefore behaves as a switch at inference: with the right type the model gains six points over the baseline, without a signal it returns to the baseline's level, and with a wrong or blended signal it collapses.

D-token differs in one respect that matters for deployment. Every perturbation collapses it, including the in-distribution class-mean condition ($+0.073$, cluster CI $[+0.040, +0.107]$), so its sensitivity is not an artifact of an out-of-distribution zero vector, and the zeroed model is worse than the flat baseline by -0.067 (cluster CI $[-0.098, -0.038]$), whereas the zeroed D-hook is not. Adaptation with a prepended token makes the model depend on the token; adaptation with an additive vector does not, even though the vector is used. In both mechanisms the conditioning acts almost entirely on clicks. Attention measured at the token that predicts the y coordinate moves the same way as accuracy under these interventions on a single seed (Appendix G); we treat those numbers as a consistency check, not as evidence about faithfulness on their own.

6.2 A positional-encoding failure that inverted a conclusion

Our first implementation of the prepended token replaced the slot's embedding by building `inputs_embeds` and passing that to the model instead of `input_ids`. On `taps_and_swipes` that variant scored 0.298 ± 0.026 hit@0.10 against 0.390 ± 0.010 for the flat baseline in the same set of runs, and a variant with the same slot but a frozen random embedding scored the same (0.290), which pointed at the injection path. Qwen2-VL derives its three-dimensional rotary positions from `input_ids`; given `inputs_embeds` alone, transformers falls back to one-dimensional positions replicated across the three axes, and the image tokens lose their two-dimensional layout (details, version, and a related library report [Hugging Face Transformers contributors, 2024] in Appendix F). Modifying embeddings through a forward hook with `input_ids` intact removed the deficit. We record this because the failure is silent and we initially read it as evidence against conditioning.

7 Limitations

The study is small: one 2B backbone, LoRA adapters, validation slices of 200 to 250 steps, one learning rate and one adapter configuration for every variant, and one L4 per run. The absolute grounding numbers are far below the field's leaderboards, per-seed variance is large, and the conclusions rest on paired statistics reported at two levels because neither is clean: the cluster bootstrap ignores that seeds share a data order, and the seed-level test has five or three units. The refutation of the prepended token is conditional on that untuned configuration; a stronger D-token could exist under a schedule that trains the fresh token faster, and we did not train a control that keeps the slot but freezes its embedding. The headline stream's only spatially informative class has a fixed off-screen target, so the claim is about adaptation on a mixed stream that contains such a class, not about a general spatial prior. The end-to-end result is descriptive, the scaling study scores each size on a different validation slice with 24 uncorrected contrasts, the attention analysis uses one seed and one layer with a probe token chosen after the fact, and Mind2Web is at floor. We do not evaluate on ScreenSpot or on multi-step task success, where the compounding-error motivation of the introduction would have to be tested directly. Code, split indices, and prompts will be released.

8 Conclusion

Supervising the action type during adaptation of a small grounding model helps when the training stream mixes groundable actions with a class whose target is degenerate; the help is concentrated on clicks, it is protection from that class rather than a spatial prior, and without such a class conditioning does nothing. An auxiliary loss, an additive embedding, and the type written into the prompt capture the gain; the prepended learned token we set out to validate is consulted by the model and improves nothing, and unlike the additive embedding it leaves a model that grounds below the baseline when the signal is removed. The lessons for adapting small VLMs on mixed action streams are to supervise the action type, to prefer conditioning that degrades to the baseline rather than below it, and to keep `input_ids` on the path whenever conditioning touches the embeddings.

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A Training and evaluation details

Data. AITW screenshots in the mirror we use are stored as raw RGB byte buffers without an image header; we decode them by matching the buffer length against seven observed resolutions (for example 540×1140 and 412×732) with a portrait-aspect fallback. Mind2Web screenshots are downscaled so that no image exceeds one megapixel. Examples are drawn from the AITW training stream in stored order, filtered to the mix's action labels; the first n eligible steps form the training slice and the next 250 (200 for the control mix) form validation. Consecutive steps belong to the same episodes, so a validation slice spans a few dozen episodes and may share its boundary episode with training. Table 8 lists the class mix of each validation slice.

Prompts. Stage 2 prompts contain the screenshot, the line `Goal: {goal}`, and the instruction `Predict the action coordinate.` (C uses `Predict the action type and coordinate.` and is trained to answer with the action word before the coordinate; D-text prefixes `Action: {word}.`). D-token places <|action_slot|> at the start of the text; exactly one slot per sequence is asserted at every step.

Optimization. AdamW with learning rate 2×10^{-5} and weight decay 0.01, batch size 1, gradient clipping at 1.0, two epochs, a fixed per-epoch shuffle seeded by the run seed. The action-embedding table (D-hook, D-token) and the auxiliary head (B) share the base learning rate. Evaluation is greedy with at most 16 new tokens; the coordinate is parsed with a regular expression and an unparseable output counts as a miss at distance $\sqrt{2}$. Greedy evaluation is deterministic for a trained model; models in different tables that share a seed are separately trained copies (Table 9 and the intervention runs retrain the variant), and their numbers differ by training nondeterminism.

Compute. Everything runs on single NVIDIA L4 GPUs. A Stage-2 run at 1,200 training examples takes about 35 minutes; the full study, including retracted and re-run cells, cost about 130 GPU-hours.

Statistics. For each contrast we align the per-example distances of the two variants on every shared seed. The cluster bootstrap resamples the validation examples with replacement and keeps all seeds of a sampled example together, recomputes the mean difference 10,000 times, and reports the percentile interval and the two-sided bootstrap p. The seed-level test is a paired t-test on the per-seed means. The earlier (seed, example)-unit bootstrap and its permutation counterpart are reported in the released logs; their intervals are narrower by about a third and every star in this paper was recomputed on the cluster basis.

B Stage 1, Mind2Web, and per-class tables

Table 5: Stage 1 action-type macro-F1 (three seeds). On Mind2Web the instruction names the action, so the screenshot adds little; on AITW the instruction describes a goal and the screenshot is what separates the classes.

Features	Mind2Web (2 classes)	AITW (5 classes)
text only	0.597 ± 0.011	0.141 ± 0.032
text + screenshot	0.605 ± 0.016	0.555 ± 0.036
zeroed (sanity)	0.469 ± 0.000	0.110 ± 0.051
TF-IDF + logistic regression	0.622	0.168
majority class	0.472	0.140

Table 6: Multimodal-Mind2Web grounding control (1,000 training and 250 evaluation steps, three seeds, seed-level only). The target is the center of the gold element; hit@bbox is the fraction of predicted points inside the element. Both models are at floor.

Variant	hit@0.10	hit@0.25	hit@bbox	mean L2 $\downarrow$
A: flat	0.365 ± 0.031	0.645 ± 0.035	0.095 ± 0.018	0.234 ± 0.023
D-hook: additive	0.355 ± 0.032	0.664 ± 0.052	0.104 ± 0.034	0.233 ± 0.027

Table 7: Removing *type* events from the training slice only, scored on the identical headline validation slice (three seeds). Deltas are paired cluster bootstraps over examples; seed-level p in parentheses.

Variant	Training stream	hit@0.10	click hit@0.10	scroll hit@0.10	Δhit@0.10 vs. A (same stream)
A: flat	with type	0.255 ± 0.026	0.294 ± 0.042	0.304 ± 0.022	
A: flat	type removed	0.297 ± 0.024	0.365 ± 0.039	0.275 ± 0.013	
B: aux. loss	with type	0.305 ± 0.015	0.373 ± 0.020	0.290 ± 0.013	
B: aux. loss	type removed	0.319 ± 0.034	0.404 ± 0.063	0.246 ± 0.066	+0.021 [-0.004, +0.045] (0.391)
D-hook: additive	with type	0.300 ± 0.035	0.351 ± 0.048	0.341 ± 0.045	
D-hook: additive	type removed	0.305 ± 0.050	0.357 ± 0.074	0.348 ± 0.000	+0.008 [-0.020, +0.035] (0.858)

C Type events removed from training

D Full scaling table

E The earlier three-condition D-token test

Table 9 is the test that preceded Table 4: separately trained D-token models (five seeds), decoded with the gold embedding, the cyclic wrong map, and the table zeroed. Its numbers differ from the D-token rows of Table 4 by training nondeterminism and agree in every conclusion.

F The `inputs_embeds` failure in detail

Qwen2-VL locates the image tokens in `input_ids` to assign each visual token a temporal, height, and width position for its three-dimensional rotary embedding. When a caller supplies `inputs_embeds` and no `input_ids`, the model cannot perform that lookup; in the transformers releases our runs used (pinned at 4.45 or later; the current 5.9 release behaves the same) the positions become a one-dimensional sequence index replicated across the three axes, so every image token is positioned as text and the two-dimensional layout the grounding task depends on is lost. A caller on that path must also merge the visual features into the embeddings itself, a second opportunity for silent error. We inferred the cause from the code and from the frozen-embedding control described in Section 6.2 rather than by dumping position ids at the time. The same dependence on `input_ids` was reported as a generation crash in December 2024 [Hugging Face Transformers contributors, 2024]; the silent degradation when images are present is the case a fine-tuning practitioner meets. In the set of runs where the failure appeared, the fixed additive variant scored 0.392 ± 0.043 hit@0.10 on `taps_and_swipes` against the flat baseline's 0.390 ± 0.010; the re-run of that control with per-example logging (Table 3) gives 0.363 ± 0.021 against 0.382 ± 0.015.

G Where the model looks

We measured where the models look, with the caveats that the measure is head-averaged attention at one layer, that attention is not information flow, and that teacher-forcing the gold answer places the gold x coordinate in the prefix of the token that predicts y. Under gold conditioning both embedding models put 10 to 11% of that token's image attention within 0.10 of the target, against a chance share of 2.6%; the last prompt token, which our earlier visualizations used, puts about 4% there under every conditioning, so most of the localization we can see happens after the x coordinate is in the prefix. Teacher-forcing the model's own answer instead of the gold one, so that the y-predicting token sees the predicted x, lowers the target share only slightly (D-hook 0.092, D-token 0.085), so

Table 8: Conditioning advantage vs. training-set size (AITW `all_with_coords`, 3 seeds per cell). Intervals are 95% cluster bootstraps over validation examples (all seeds of a sampled example kept together); stars from the same bootstrap. Absolute values are not comparable across rows because the validation slice moves with n; its click/scroll/type counts are given per row. The study is descriptive: 24 contrasts, no multiplicity correction.

n_{train}	val mix (c/s/t)	A hit@0.10	B − A	D-hook − A	B − A (hit@0.25)	D-hook − A (hit@0.25)
300	170/33/47	0.139 ± 0.032	+0.067 [+0.031, +0.103]***	+0.069 [+0.041, +0.099]***	-0.055 [-0.092, -0.019]**	+0.067 [+0.033, +0.101]***
500	198/12/40	0.315 ± 0.059	+0.011 [-0.019, +0.040]	+0.001 [-0.032, +0.033]	+0.004 [-0.027, +0.033]	+0.059 [+0.025, +0.092]***
800	141/71/38	0.261 ± 0.020	-0.040 [-0.073, -0.008]*	+0.048 [+0.009, +0.087]*	-0.039 [-0.069, -0.008]*	+0.064 [+0.025, +0.103]***
1200	169/46/35	0.255 ± 0.026	+0.051 [+0.021, +0.081]***	+0.045 [+0.015, +0.079]**	+0.071 [+0.036, +0.107]***	+0.055 [+0.017, +0.092]**
2500	166/53/31	0.244 ± 0.037	-0.012 [-0.039, +0.015]	+0.037 [+0.009, +0.067]**	-0.008 [-0.043, +0.028]	+0.104 [+0.067, +0.143]***
5000	187/29/34	0.363 ± 0.009	+0.005 [-0.025, +0.037]	+0.043 [+0.011, +0.076]**	-0.023 [-0.051, +0.005]	+0.032 [+0.001, +0.064]*

Table 9: D-token causal-use test: the same trained model evaluated with the gold action embedding, a wrong (cyclically permuted) one, and a zeroed one (5 seeds, paired bootstrap over pooled examples).

Conditioning	hit@0.05	hit@0.10	hit@0.25	mean L2 ↓
gold	0.140 ± 0.059	0.236 ± 0.058	0.502 ± 0.045	0.386 ± 0.020
wrong	0.014 ± 0.004	0.079 ± 0.013	0.200 ± 0.027	0.728 ± 0.081
zero	0.074 ± 0.055	0.167 ± 0.066	0.416 ± 0.103	0.489 ± 0.059
gold − wrong		+0.157 [+0.114, +0.201]***	+0.302 [+0.246, +0.356]***	-0.342 [-0.375, -0.309]***
gold − zero		+0.069 [+0.038, +0.100]***	+0.086 [+0.056, +0.117]***	-0.103 [-0.121, -0.086]***

the localization is not an artifact of the supplied x. The interventions move attention the same way they move accuracy on this seed (Figure 3, Table 10): a wrong type roughly halves the target share for D-hook, and zeroing costs D-token both attention and accuracy. On this one seed zeroing left D-hook's attention and accuracy unchanged, which the five-seed study in Section 6.1 shows to be within the noise of a small real cost. Table 10 gives the target-attention share for the last prompt token (the one that predicts the opening parenthesis), the token that predicts the first x digit, and the token that predicts the first y digit.

Table 10: Share of image attention within 0.10 of the target (chance 0.026) at three probe positions, seed 42, 120 screens, 3/4-depth layer, gold answer teacher-forced.

Model	Position	gold	wrong (cyclic)	wrong (click↔scroll)	zero
D-hook	last prompt token	0.040	0.036	0.035	0.036
D-hook	pre-x token	0.039	0.036	0.037	0.039
D-hook	pre-y token	0.111	0.064	0.049	0.119
D-token	last prompt token	0.038	0.034	0.034	0.035
D-token	pre-x token	0.038	0.034	0.033	0.035
D-token	pre-y token	0.095	0.080	0.079	0.079

H Qualitative examples

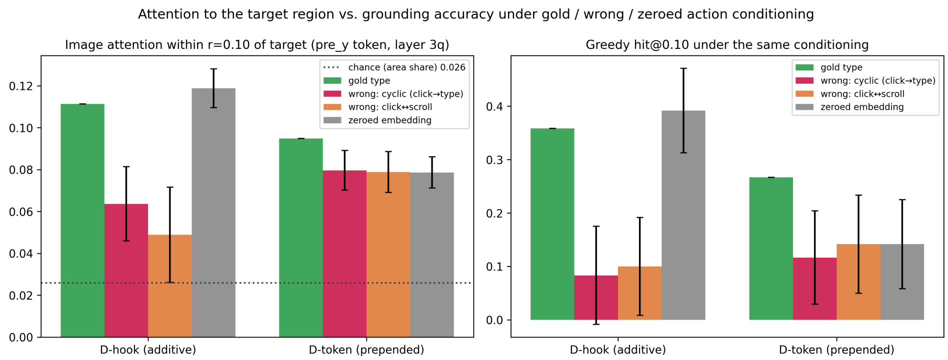

Figure 3: Same trained model, four conditionings, 120 validation screens, one seed per variant. Left: share of image attention within 0.10 of the target from the token that predicts the y coordinate, with the gold answer teacher-forced (3/4-depth layer, head-averaged); the dotted line is the share of image tokens within that radius. Right: greedy hit@0.10 under the same conditioning. Error bars are 95% paired-bootstrap intervals of the difference from gold over the 120 screens.

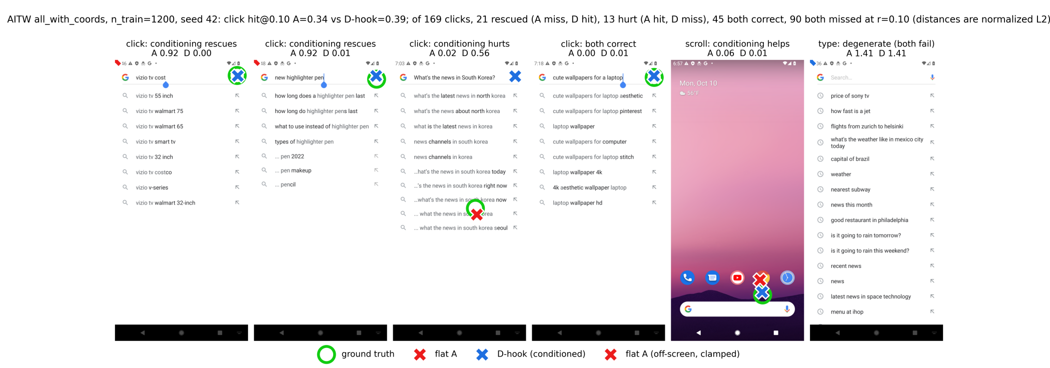

Figure 4: Predictions of the flat baseline (red) and D-hook (blue) against the target (green) on held-out AITW screens at the headline setting, one seed. The panels are a fixed spread of outcomes rather than a selection of wins: two rescues, a case where conditioning hurts, a case where both are right, a scroll, and a degenerate *type* example on which both models miss at distance $\sqrt{2}$. The title reports the base rates on this seed at $r = 0.10$: of 169 clicks, 21 are rescued by conditioning, 13 are hurt, 45 are correct under both models, and 90 are missed by both. In the rescue panels the baseline's prediction is off-screen and drawn clamped at the corner.